

Semester 1 · Academic Year 2026–2027

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Exercise 3. Systems and equilibrium [1.A · 1.B]

(3 marks)

A weather balloon (sealed, flexible envelope) is:

- | | |
|-----------------------------------|--|
| ☐ open | ☐ closed but not isolated |
| ☐ isolated | ☐ neither closed nor open |

A gas in an insulating cylinder closed by a rigid lid is:

- | | |
|--|---|
| ☐ isolated | ☐ open |
| ☐ closed but not isolated | ☐ necessarily at 0 K |

Thermodynamic equilibrium requires:

- ☐ only a steady state;
- ☐ only homogeneity;
- ☐ both: p and T uniform in space and constant in time, and equal to those of the surroundings;
- ☐ only thermal equilibrium.

Exercise 4. State diagrams and coefficients [1.C · 2.B]

(4 marks)

The slope of an isotherm drawn as V versus p is:

- | | | | |
|---|---|---|--------------------------------|
| ☐ $\left(\frac{\partial V}{\partial p}\right)_T$ | ☐ $\left(\frac{\partial V}{\partial T}\right)_p$ | ☐ $\left(\frac{\partial p}{\partial T}\right)_V$ | ☐ V/p |
|---|---|---|--------------------------------|

For a gas, the sign of $\left(\frac{\partial V}{\partial T}\right)_p$ is:

- | | | | |
|-----------------------------------|-----------------------------------|-------------------------------|---------------------------------------|
| ☐ positive | ☐ negative | ☐ zero | ☐ undetermined |
|-----------------------------------|-----------------------------------|-------------------------------|---------------------------------------|

The piezothermic coefficient is defined by:

- | | | | |
|---|---|--|---|
| ☐ $\frac{1}{p} \left(\frac{\partial p}{\partial T}\right)_V$ | ☐ $\frac{1}{V} \left(\frac{\partial V}{\partial T}\right)_p$ | ☐ $-\frac{1}{V} \left(\frac{\partial V}{\partial p}\right)_T$ | ☐ $\frac{1}{T} \left(\frac{\partial T}{\partial p}\right)_V$ |
|---|---|--|---|

The SI unit of χ_T is:

- | | | | |
|---|--|--------------------------------------|----------------------------------|
| ☐ Pa^{-1} | ☐ K^{-1} | ☐ Pa | ☐ no unit |
|---|--|--------------------------------------|----------------------------------|

[3.A]

(5 marks)

(a) An aerosol can (rigid) contains a gas at p_1 and θ_1 . It is left in the sun and reaches θ_2 . Which law applies? Compute the final pressure p_2 . (b) The bubble of a Cartesian diver, of volume V at pressure p , is compressed at constant temperature to the pressure p' . Compute its new volume V' .

(a) $p_1 = 3.0 \text{ bar}$	$\theta_1 = 27^\circ\text{C}$	$\theta_2 = 127^\circ\text{C}$
(b) $V = 1.0 \text{ cm}^3$	$p = 1.0 \text{ bar}$	$p' = 1.25 \text{ bar}$

Marking

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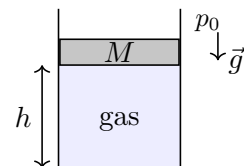
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[1.B · 3.B]

(7 marks)

A vertical cylinder of cross-section S is closed by a frictionless piston of mass M . The gas below it, treated as an ideal gas, fills a height h_1 at temperature T_1 . Outside, the pressure is p_0 .

(a) Find the pressure p of the gas. (b) Find the amount of gas n . (c) The gas is heated slowly to T_2 , the piston remaining free. Find the new height h_2 and the displacement of the piston.


$$S = 100 \text{ cm}^2 \quad M = 20 \text{ kg} \quad g = 10 \text{ m s}^{-2} \quad p_0 = 1.0 \times 10^5 \text{ Pa}$$

$$h_1 = 25 \text{ cm} \quad T_1 = 300 \text{ K} \quad T_2 = 360 \text{ K} \quad RT_1 \approx 2.5 \times 10^3 \text{ J mol}^{-1}$$

Marking

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